Part 3: Inferential Statistics

From observed patterns to uncertainty and business decisions · 307312 Applied Statistics

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- **Sequence:** Probability distributions → estimation → hypothesis testing → regression.
- **Superstore:** learn probability and sampling from a known finite set of retail orders.
- **COLA:** compare weekly sales in two display conditions using a randomized business experiment.
- **Consumer:** model credit-card charges using household income and household size.
- Report the estimated effect, uncertainty, assumptions and business relevance together.

3.1 Three questions and three designs

Question	Dataset and observational unit	What supports the analysis?
How accurately can a sample estimate average order sales?	Superstore: one order	We explicitly draw a random sample from the fixed course frame
Do display conditions differ in mean weekly cola sales?	COLA: one store observed for one week	The book describes random assignment of 20 selected stores to two display conditions
How do household characteristics relate to card charges?	Consumer: one consumer record	An observational regression model; inference is conditional on model and sampling assumptions

- Superstore is fictitious; its data do not establish facts about real retailers.
- An independent row is a design assumption, not a property guaranteed by a spreadsheet.
- COLA has different stores in its two columns; aligned rows are **not pairs**.
- Consumer provenance does not establish random selection or causal effects.

```
# Each import cell locates data independently of YAML or earlier notebooks.
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
cola <- read.csv(file.path(data_dir, "business", "cola.csv"))
consumer <- read.csv(file.path(data_dir, "business", "consumer.csv"))
knitr::kable(data.frame(data = c("Superstore orders", "COLA stores", "Consumer records"),
                        n = c(nrow(orders), nrow(cola), nrow(consumer))))
```

data	n
Superstore orders	3322
COLA stores	20
Consumer records	50

3.2 Probability starts with a defined experiment

Experiment: Choose one of the 3,322 course-frame orders with equal probability.

- Sample space: all course-frame order IDs.
- Event R: order ID appears in the source Returns table.
- Event L: recorded total order profit is negative.
- $0 \le P(A) \le 1; P(S) = 1; P(A^c) = 1 - P(A)$.
- For disjoint events, probabilities add.

```
probabilities <- orders %>% summarise(
  p_listed_return = mean(listed_return), p_loss_order = mean(loss_order),
  both = mean(listed_return & loss_order),
  either = mean(listed_return | loss_order))
knitr::kable(probabilities, digits = 4)
```

p_listed_return	p_loss_order	both	either
0.0575	0.202	0.0108	0.2486

- These are exact proportions for the fixed frame, interpreted as probabilities under uniform random order selection.
- "Not listed" means absent from the supplied Returns table; it does not independently verify that no return occurred.
- Do not subtract presumed refunds from recorded profit; the workbook does not establish how returns enter the accounting fields.

Conditional probability and independence

$$P(R \mid L) = \frac{P(R \cap L)}{P(L)}, \qquad P(R \cup L) = P(R) + P(L) - P(R \cap L).$$

```
c(p_return = mean(orders$listed_return),
  p_return_given_loss = mean(orders$listed_return[orders$loss_order]),
  p_return_given_no_loss = mean(orders$listed_return[!orders$loss_order]))
```

```
            p_return    p_return_given_loss p_return_given_no_loss
          0.05749548             0.05365127             0.05846850
```

- Conditioning changes the denominator.
- Independence requires $P(R \cap L) = P(R)P(L)$, or equivalent conditional relationships.
- Different conditional rates in this frame describe an association, not a causal effect.

Reverse the conditioning with Bayes' rule

$$P(L \mid R) = \frac{P(R \mid L)P(L)}{P(R)}.$$

```
p_l <- mean(orders$loss_order)
p_r <- mean(orders$listed_return)
p_r_given_l <- mean(orders$listed_return[orders$loss_order])
c(direct = mean(orders$loss_order[orders$listed_return]),
  bayes = p_r_given_l * p_l / p_r)
```

```
   direct     bayes
0.1884817 0.1884817
```

- The two answers agree because they use the same events and denominator definitions.
- $P(L \mid R)$ and $P(R \mid L)$ answer different business questions.

3.3 Random variables and distributions

- A random variable assigns a number to the outcome of an experiment.
- A **discrete** variable has countable values, such as the number of listed-return orders in a sample.
- A **continuous model** describes values on a continuum, such as a model for measurement error.
- For discrete X, $\sum_x P(X = x) = 1$ and $E(X) = \sum_x xP(X = x)$.
- $\text{Var}(X) = E[(X - E(X))^2]$ measures model variation.

Binomial: a model with stated assumptions

Model question: If ten independent future orders each had listed-return probability p, what is the chance of at least one listed return?

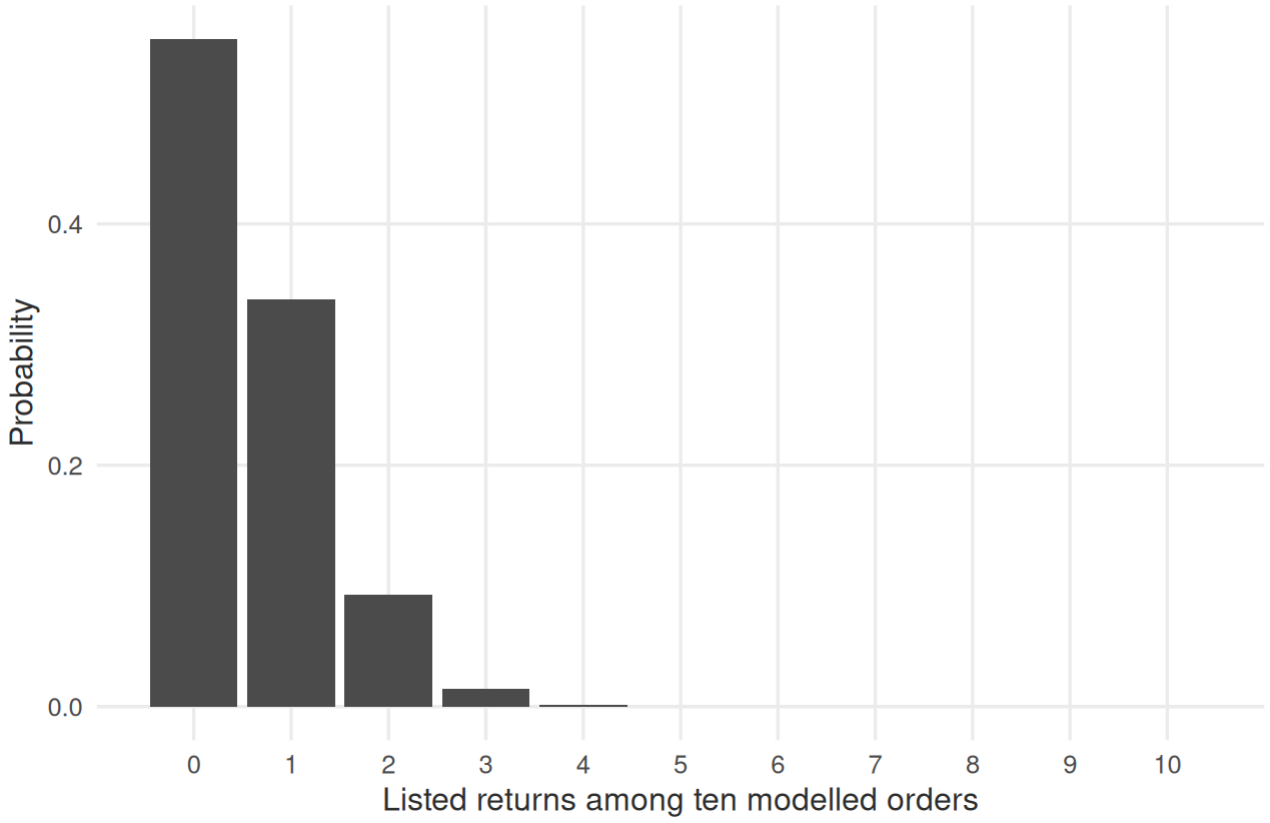
- Fixed number of trials; two outcomes per trial.
- Independent trials; constant p.
- Using the historical frame proportion as p is a **model assumption**, not a verified forecast.

$$X \sim \text{Binomial}(n, p), \quad E(X) = np, \quad \text{Var}(X) = np(1 - p).$$

```
p <- mean(orders$listed_return)
c(p_at_least_one = 1 - pbinom(0, size = 10, prob = p),
  expected_count = 10 * p, count_sd = sqrt(10 * p * (1 - p)))
```

```
p_at_least_one expected_count       count_sd
     0.4468609      0.5749548      0.7361369
```

```
binomial <- data.frame(returns = 0:10, probability = dbinom(0:10, 10, p))
ggplot(binomial, aes(returns, probability)) +
  geom_col(fill = petra_ink) + scale_x_continuous(breaks = 0:10) +
  labs(x = "Listed returns among ten modelled orders", y = "Probability")
```



Hypergeometric: sampling the known frame without replacement

Different experiment: Randomly select ten distinct course-frame orders.

```
N <- nrow(orders)
K <- sum(orders$listed_return)
c(exact_without_replacement = 1 - dhyper(0, m = K, n = N - K, k = 10),
  binomial_approximation = 1 - dbinom(0, size = 10, prob = K / N))
```

```
exact_without_replacement    binomial_approximation
                0.4473186                 0.4468609
```

- Hypergeometric probabilities account for sampling without replacement.
- The binomial is close here because ten orders are a small fraction of the frame.
- Sampling product lines instead would overweight orders containing many lines.

Poisson: counts over an exposure period

- A Poisson model assumes independent events and a constant rate over the chosen exposure.
- The following rate, **three orders per hour**, is an illustrative planning assumption.
- It is not estimated from Superstore: the file has dates, not order times.

```
c(p_more_than_five = 1 - ppois(5, lambda = 3),
  mean_count = 3, variance_count = 3)
```

```
p_more_than_five       mean_count   variance_count
      0.08391794       3.00000000       3.00000000
```

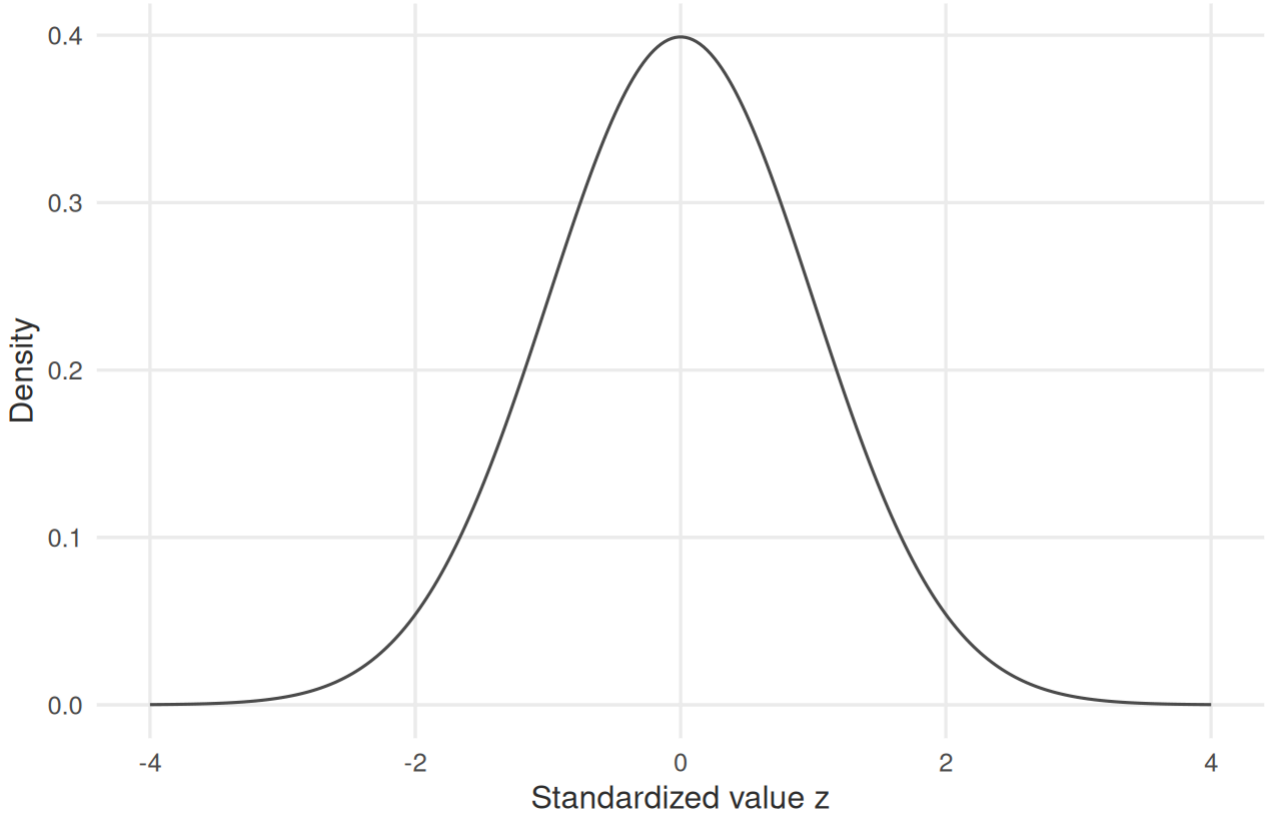
Normal distributions and standardization

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1) \quad \text{when } X \sim N(\mu, \sigma^2).$$

```
c(p_within_two_sd = pnorm(2) - pnorm(-2),
  upper_975_percentile = qnorm(.975))
```

```
    p_within_two_sd upper_975_percentile
          0.9544997            1.9599640
```

```
normal_curve <- data.frame(z = seq(-4, 4, length.out = 300))
ggplot(normal_curve, aes(z, dnorm(z))) + geom_line(colour = petra_ink) +
  labs(x = "Standardized value z", y = "Density")
```



- Area under the curve gives probability; height at a point is a density.
- For a continuous model, $P(X = x) = 0$ at any exact point.
- Superstore order sales are strongly skewed; we do not assume the raw values are normal.

3.4 Sampling distributions and standard errors

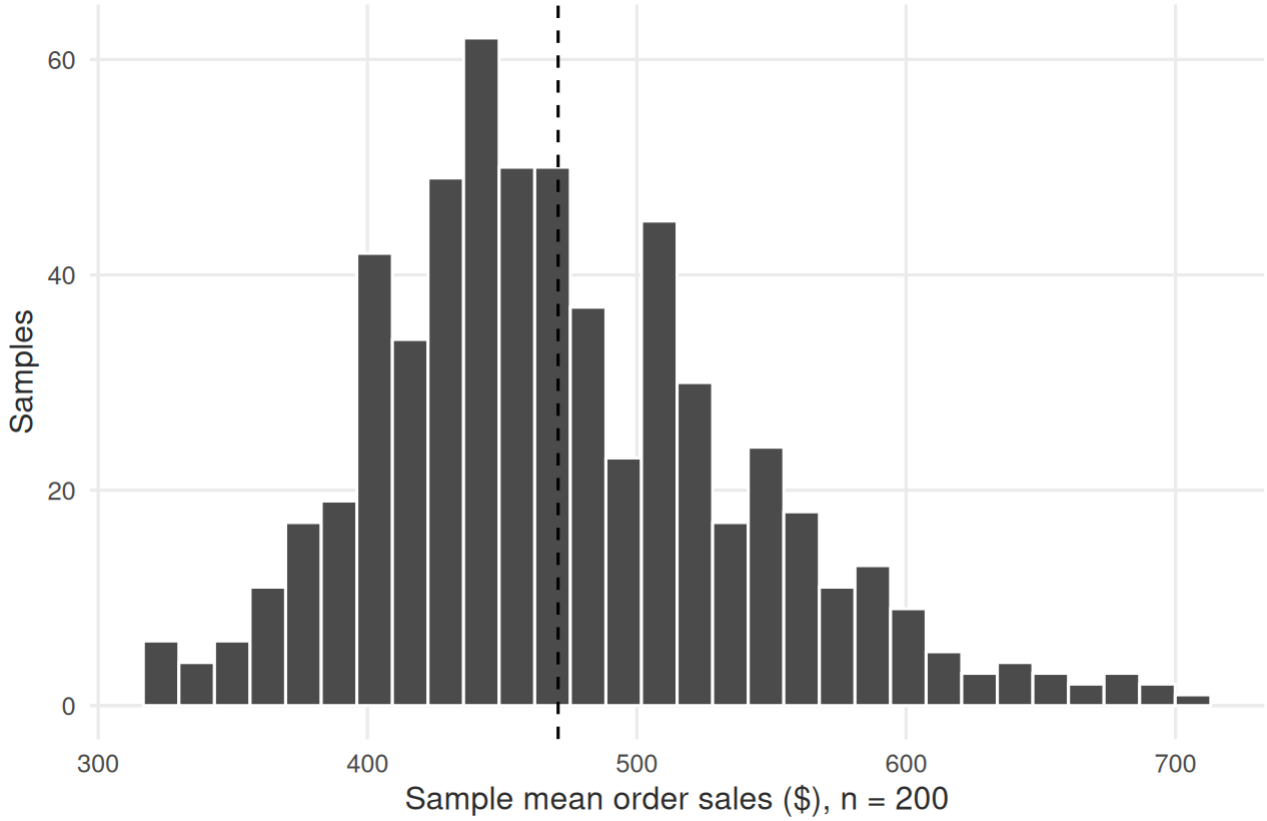
- A **parameter** describes the target population or frame.
- A **statistic** is calculated from a sample.
- **SD:** variation between individual orders.
- **SE:** variation of an estimator across repeated samples.
- Under independent sampling, $SE(\bar{x}) \approx s/\sqrt{n}$.
- Without replacement from a finite frame, multiply by $\sqrt{1 - n/N}$ when using the sample variance with denominator $n - 1$.

Repeated sampling from Superstore

```
set.seed(307312)
n_sample <- 200
B <- 600
sample_means <- replicate(B, mean(sample(orders$sales, n_sample, replace = FALSE)))
knitr::kable(data.frame(frame_mean = mean(orders$sales),
  mean_of_sample_means = mean(sample_means),
  empirical_se = sd(sample_means),
  design_se = sd(orders$sales) / sqrt(n_sample) * sqrt(1 - n_sample / N)), digits = 2)
```

frame_mean	mean_of_sample_means	empirical_se	design_se
470.8	471.94	68.83	67.38

```
ggplot(data.frame(sample_mean = sample_means), aes(sample_mean)) +
  geom_histogram(bins = 30, fill = petra_ink, colour = "white") +
  geom_vline(xintercept = mean(orders$sales), linetype = "dashed") +
  labs(x = "Sample mean order sales ($), n = 200", y = "Samples")
```



- The dashed line marks the known frame mean.
- Sample means vary even though the source data do not change.
- The central limit theorem concerns the sampling distribution of a mean, not normality of the raw data.
- Severe skewness may require a large sample; “30 observations” is not a universal guarantee.
- The randomization we perform supports inference to this **fixed frame**, despite repeated customers in the frame. It does not justify independent future-order models.

3.5 Point and interval estimation

Question: Can a random sample estimate average sales across all course-frame orders?

```
set.seed(20261106)
sampled_orders <- orders[sample.int(N, n_sample, replace = FALSE), ]
xbar <- mean(sampled_orders$sales)
se <- sd(sampled_orders$sales) / sqrt(n_sample) * sqrt(1 - n_sample / N)
critical <- qt(.975, df = n_sample - 1)
mean_ci <- xbar + c(-1, 1) * critical * se
knitr::kable(data.frame(sample_n = n_sample, estimate = xbar,
  lower_95 = mean_ci[1], upper_95 = mean_ci[2], frame_mean = mean(orders$sales)), digits = 2)
```

sample_n	estimate	lower_95	upper_95	frame_mean
200	552.81	389.98	715.64	470.8

- This is an **approximate** t interval with a finite-frame correction.
- The target is the mean order sales in the fixed US, 2023–2025 course frame.
- In repeated sampling, a 95% procedure aims to cover the fixed target about 95% of the time.
- After the interval is computed, it either contains the target or it does not.
- It does not mean that 95% of individual orders lie inside it.

See coverage rather than assume it

```
set.seed(907)
intervals <- replicate(600, {
  z <- sample(orders$sales, n_sample, replace = FALSE)
  half_width <- qt(.975, n_sample - 1) * sd(z) / sqrt(n_sample) * sqrt(1 - n_sample / N)
  mean(z) + c(-1, 1) * half_width
})
c(simulated_coverage = mean(intervals[1, ] <= mean(orders$sales) &
                            intervals[2, ] >= mean(orders$sales)),
  mean_interval_width = mean(intervals[2, ] - intervals[1, ]))
```

```
simulated_coverage mean_interval_width
         0.8833333         244.6788357
```

- Simulation results fluctuate; 600 runs do not establish the exact coverage probability.
- In this seeded run, coverage is about **88%**, materially below the nominal 95%. With this skewed frame, 200 orders are insufficient for reliable nominal t coverage.
- This is a diagnostic counterexample: do not treat the nominal 95% label as validated precision. The full frame is available, so its exact mean is the primary descriptive answer.
- Investigate a larger sample and compare empirical coverage before relying on this interval in a sampling project.
- Increasing confidence widens an interval; increasing sample size usually narrows it.

```r
# Compare the same approximate procedure at several sampling fractions.
coverage_by_size <- do.call(rbind, lapply(c(200, 1000, 2000), function(n) {
  set.seed(907)
  bounds <- replicate(600, {
    z <- sample(orders$sales, n, replace = FALSE)
    h <- qt(.975, n - 1) * sd(z) / sqrt(n) * sqrt(1 - n / N)
    mean(z) + c(-1, 1) * h
  })
  data.frame(sample_n = n, fraction = n / N,
    empirical_coverage = mean(bounds[1, ] <= mean(orders$sales) &
                              bounds[2, ] >= mean(orders$sales)))
}))
knitr::kable(coverage_by_size, digits = 3)
```

sample_n	fraction	empirical_coverage
200	0.060	0.883
1000	0.301	0.933
2000	0.602	0.933

- Larger samples improve coverage here, at a higher collection cost.
- This is evidence about this frame and procedure, not a universal sample-size rule.

Estimate a proportion

```r
return_interval <- prop.test(sum(sampled_orders$listed_return), n_sample,
                             conf.level = .95, correct = FALSE)
c(estimate = unname(return_interval$estimate),
  lower_95 = return_interval$conf.int[1], upper_95 = return_interval$conf.int[2],
  frame_proportion = mean(orders$listed_return))
```

```
      estimate       lower_95       upper_95 frame_proportion
    0.06000000     0.03465219     0.10193169     0.05749548
```

- This score interval uses a binomial approximation and does not apply the finite-frame correction.
- Our sample is about 6% of the frame; an exact without-replacement analysis would use the hypergeometric distribution.
- The result describes the share **listed in the Returns table**, not a verified real-world return probability.

```r
# An exact finite-frame alternative inverts the hypergeometric tails.
observed_returns <- sum(sampled_orders$listed_return)
possible_totals <- 0:N
compatible <- phyper(observed_returns - 1, possible_totals, N - possible_totals,
                     n_sample, lower.tail = FALSE) >= .025 &
              phyper(observed_returns, possible_totals, N - possible_totals,
                     n_sample) >= .025
range(possible_totals[compatible]) / N
```

```
[1] 0.03220951 0.10114389
```

- These bounds concern the unknown fraction in a finite frame of 3,322 orders under simple random sampling without replacement.
- Discreteness can make exact coverage conservative. We can check against the known full-frame fraction in this exercise.

Sample-size planning

For a large-population mean with a planning SD s_0 and desired margin E:

$$n_0 \approx \left(\frac{z_{.975} s_0}{E}\right)^2.$$

```
# A $100 margin is an illustrative management precision target.
E <- 100
s0 <- sd(orders$sales)
n0 <- (qnorm(.975) * s0 / E)^2
c(large_population_n = ceiling(n0),
  finite_frame_n = ceiling(n0 / (1 + (n0 - 1) / N)))
```

```
large_population_n     finite_frame_n
               372                334
```

- This is a normal-approximation planning calculation, not a promise of realized precision.
- Real projects need a defensible pilot SD, collection cost and sampling design.

3.6 Hypothesis testing: start with the business claim

1. Define the parameter, target and design.
2. State H_0 and H_1 before inspecting results.
3. Choose the significance level and test; check assumptions.
4. Calculate the effect, interval, test statistic and p-value.
5. State what the evidence supports and whether the effect matters.

- **Type I error:** rejecting a true null; controlled at the chosen significance level under the test assumptions.
- **Type II error:** failing to reject a false null.
- **Power:** probability of rejecting a specified false null; depends on effect, noise, sample size and test.
- A p-value is a probability of a result at least as extreme **under the null model**.
- It is not the probability that the null is true or that a result occurred “by chance.”
- Failure to reject is not proof of equality or equivalence.

A one-mean test with a stated benchmark

Planning question: Is mean weekly sales under the normal display different from **50 cases**?

- Fifty cases is a chosen planning benchmark, not a publisher-established target.
- $H_0: \mu_{\text{normal}} = 50; H_1: \mu_{\text{normal}} \neq 50.$
- The following one-sample t analysis is model-based for comparable normal-display stores.
- The 10 stores were not documented as a random sample of all supermarkets.

```
normal_sales <- cola$weekly_cases[cola$display == "Normal"]
t.test(normal_sales, mu = 50, alternative = "two.sided")
```

```
    One Sample t-test

data:  normal_sales
t = 0.05066, df = 9, p-value = 0.9607
alternative hypothesis: true mean is not equal to 50
95 percent confidence interval:
 36.90395 63.69605
sample estimates:
mean of x
     50.3
```

- Use t when population SD is unknown and estimated from the sample.
- A z test for a mean requires a defensibly known population SD or a justified large-sample approximation.
- With ten observations, strong skewness or outliers would make ordinary t inference less secure.

3.7 COLA: compare display conditions

The business experiment

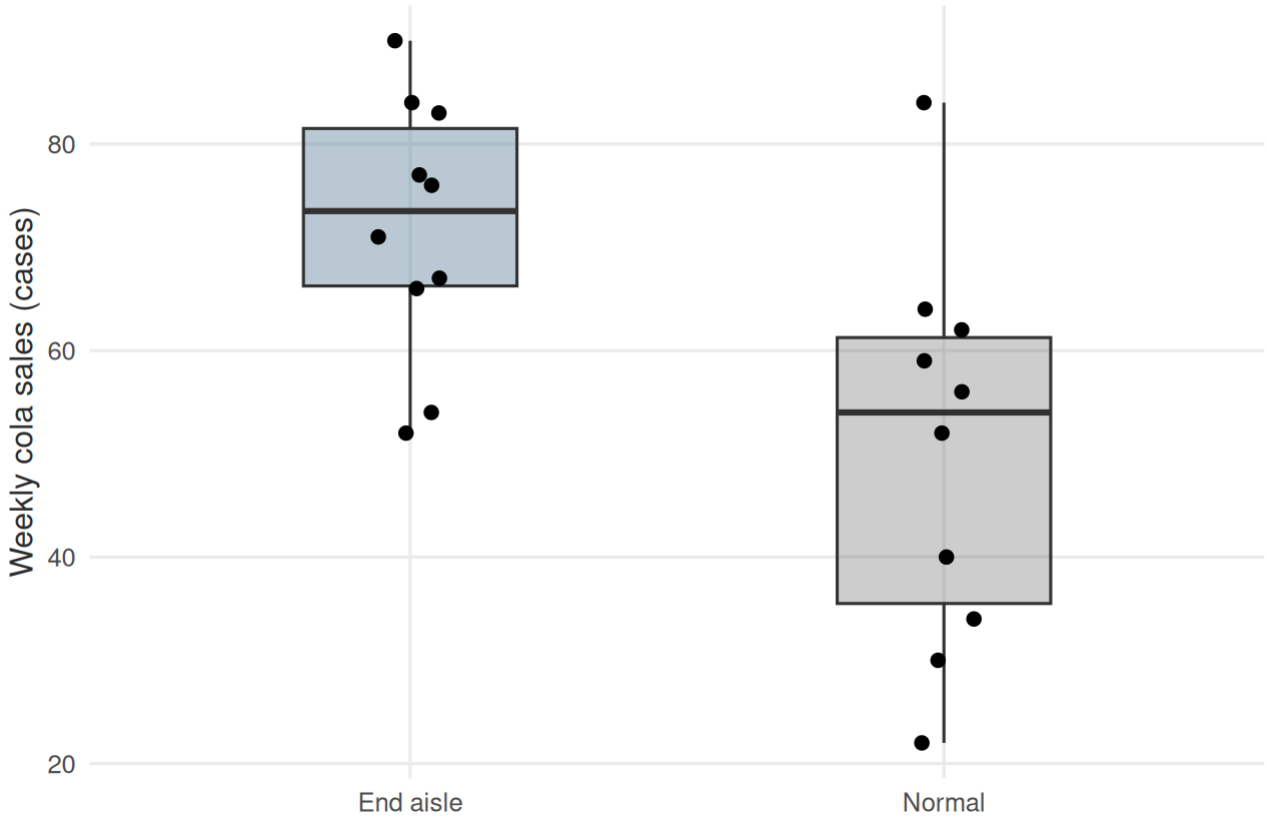
- Source: Berenson, Levine and Krehbiel, *Basic Business Statistics*, ninth edition, pp. 336–339.
- Select 20 stores with similar storewide sales volume.
- Randomly assign ten stores to normal shelving and ten to an end-aisle display.
- Record cola sales during one week, in **cases**.
- Each store contributes one observation. The course's store IDs are labels added for convenience.

Question: How large is the mean difference, and is it distinguishable from zero?

```
cola_summary <- cola %>% group_by(display) %>%
  summarise(n = n(), mean_cases = mean(weekly_cases), sd_cases = sd(weekly_cases),
            median_cases = median(weekly_cases), .groups = "drop")
knitr::kable(cola_summary, digits = 2)
```

display	n	mean_cases	sd_cases	median_cases
End aisle	10	72.0	12.54	73.5
Normal	10	50.3	18.73	54.0

```
ggplot(cola, aes(display, weekly_cases, fill = display)) +
  geom_boxplot(width = .4, alpha = .35, outlier.shape = NA) +
  geom_point(position = position_jitter(width = .06, height = 0, seed = 17), size = 2) +
  scale_fill_manual(values = petra_display) +
  labs(x = NULL, y = "Weekly cola sales (cases)") + guides(fill = "none")
```



Welch two-sample t test

Let $\Delta = \mu_{\text{end}} - \mu_{\text{normal}}$.

$$H_0 : \Delta = 0, \qquad H_1 : \Delta \neq 0, \qquad t = \frac{\bar{x}_{\text{end}} - \bar{x}_{\text{normal}}}{\sqrt{s^2_{\text{end}}/n_{\text{end}} + s^2_{\text{normal}}/n_{\text{normal}}}}.$$

```
end_sales <- cola$weekly_cases[cola$display == "End aisle"]
cola_test <- t.test(end_sales, normal_sales, alternative = "two.sided", var.equal = FALSE)
cola_test
```

```
    Welch Two Sample t-test

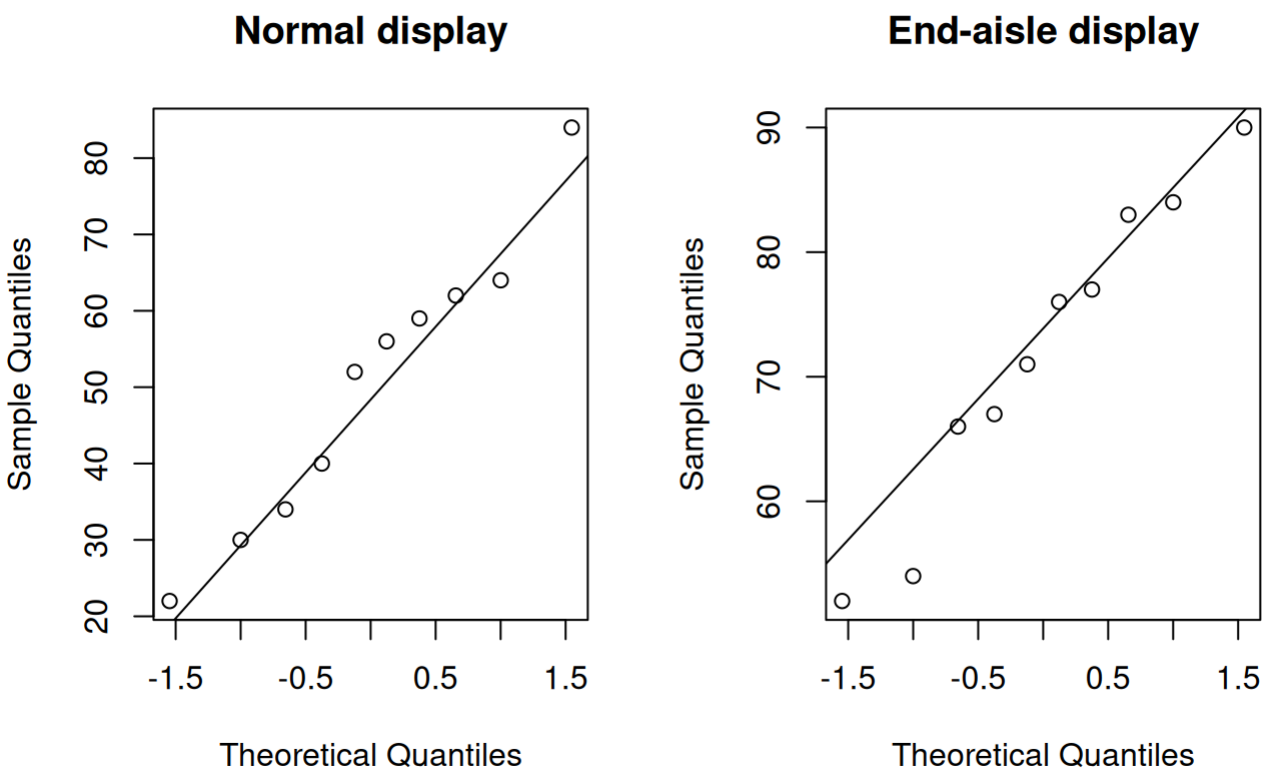
data:  end_sales and normal_sales
t = 3.0446, df = 15.723, p-value = 0.007849
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
  6.568701 36.831299
sample estimates:
mean of x mean of y
     72.0      50.3
```

- Mean difference: **21.7 additional cases per week**, end aisle minus normal.
- Approximate 95% interval: **6.57 to 36.83 cases**.
- Two-sided p-value: about **0.0078** under the Welch model.
- At $\alpha = .05$, reject the zero-difference null under the stated assumptions.
- The book uses a pooled-variance test; Welch avoids assuming equal population variances and gives slightly different results.

- Random assignment supports a causal comparison **within the experiment**, subject to implementation and no interference between stores.
- Generalizing to all stores or later weeks requires additional evidence.

Check assumptions and practical value

```
par(mfrow = c(1, 2))
qqnorm(normal_sales, main = "Normal display"); qqline(normal_sales)
qqnorm(end_sales, main = "End-aisle display"); qqline(end_sales)
```



Normal Q–Q plots by display; ten stores per group provide limited diagnostic evidence.

```
par(mfrow = c(1, 1))
```

- Check independence from the design, not a statistical normality test.
- Inspect the distributions for unusual values and strong asymmetry.
- A small-sample Q–Q plot cannot prove normality.
- More sales are not automatically more profit: obtain contribution per case, display cost and displaced-product sales.
- Do not invent these missing financial inputs.

When would a paired test be appropriate?

- If the **same stores** were measured under both conditions, analyze within-store differences.
- Account for period and carryover effects when planning such a design.
- That is not the supplied COLA experiment, so do not use a paired test on these columns.

```
# Only for a genuinely paired design with matched observations:
# t.test(after, before, paired = TRUE)
```

More than two means

```
# A one-way ANOVA expresses the same two-group comparison with equal variances.
summary(aov(weekly_cases ~ display, data = cola))
```

```
            Df Sum Sq Mean Sq F value  Pr(>F)
display      1   2354    2354   9.269 0.00697 **
Residuals   18   4572     254
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- ANOVA compares group means; with more groups, a significant F test says at least one mean differs.
- It does not identify which groups differ without planned contrasts or adjusted follow-up comparisons.
- The classical model assumes independent errors, common error variance and suitable residual behavior.
- We do not fabricate a third display condition to create an ANOVA dataset.

3.8 Categorical inference and the sampling unit

Question: In the fixed order frame, is listed-return status associated with customer segment?

```
return_by_segment <- xtabs(~ segment + listed_return, data = sampled_orders)
knitr::kable(return_by_segment)
```

	FALSE	TRUE
Consumer	96	9
Corporate	49	2
Home Office	43	1

```
expected_counts <- outer(rowSums(return_by_segment), colSums(return_by_segment)) /
  sum(return_by_segment)
knitr::kable(expected_counts, digits = 1)
```

	FALSE	TRUE
Consumer	98.7	6.3
Corporate	47.9	3.1
Home Office	41.4	2.6

```
association_test <- if (any(expected_counts < 5)) {
  fisher.test(return_by_segment)
} else {
  chisq.test(return_by_segment, correct = FALSE)
}
association_test
```

```
    Fisher's Exact Test for Count Data

data:  return_by_segment
p-value = 0.3822
alternative hypothesis: two.sided
```

- Pearson's chi-square test compares observed counts with counts expected under independence.
- Inspect expected counts; this example uses Fisher's exact test if any expected count is below five, otherwise Pearson's chi-square approximation.
- Here the random order sample supports a finite-frame exercise, with a small-sampling-fraction approximation.
- This is not evidence about future retail customers, and repeated customers limit a future-order independence model.
- Do not test all 6,682 lines as if they were independent customer responses.
- Report conditional percentages and the size of differences alongside any p-value.
- These examples are separate method demonstrations, not a preplanned confirmatory research program. Repeatedly searching for significant results inflates false positives.

3.9 Consumer: explain variation in credit-card charges

Business question and variables

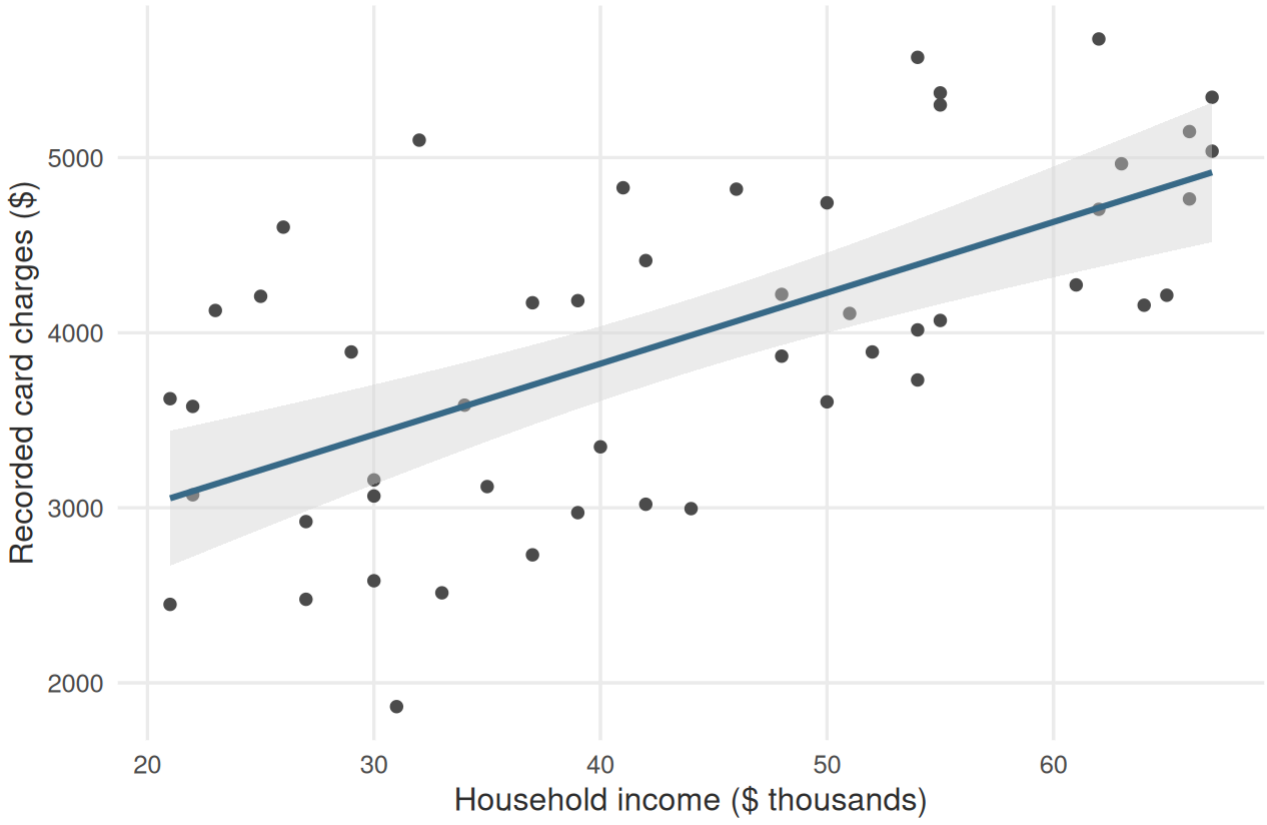
- **Question:** How are recorded card charges related to household income and household size?
- **Data:** 50 records in Anderson's `Consumer.xls` collection.
- `income_thousands`: household income in thousands of dollars.
- `household_size`: number of household members.
- `card_charges`: recorded credit-card charges in dollars.
- The workbook alone does not specify a reporting period; we do not add an annual label.
- The source is a textbook case. Authenticity, representativeness and probability sampling are not independently established.

```
knitr::kable(head(consumer))
```

income_thousands	household_size	card_charges
54	3	4016
30	2	3159
32	4	5100
50	5	4742
31	2	1864
55	2	4070

```
ggplot(consumer, aes(income_thousands, card_charges)) +
  geom_point(colour = petra_ink) +
  geom_smooth(method = "lm", se = TRUE, colour = "#3B6B8A", fill = petra_accent) +
  labs(x = "Household income ($ thousands)", y = "Recorded card charges ($)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Simple linear regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

```
simple_model <- lm(card_charges ~ income_thousands, data = consumer)
summary(simple_model)
```

```
Call:
lm(formula = card_charges ~ income_thousands, data = consumer)

Residuals:
     Min       1Q   Median       3Q      Max
-1594.87  -579.50   -66.11   511.03  1600.65

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      2204.000    329.049   6.698 2.14e-08 ***
income_thousands   40.480      7.184   5.635 9.01e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 731.7 on 48 degrees of freedom
Multiple R-squared:  0.3981,    Adjusted R-squared:  0.3856
F-statistic: 31.75 on 1 and 48 DF,  p-value: 9.012e-07
```

```
confint(simple_model)
```

```
                     2.5 %     97.5 %
(Intercept)      1542.40241 2865.59683
income_thousands   26.03566   54.92388
```

- Response Y: card charges. Predictor X: income in thousands of dollars.
- Least squares chooses the line minimizing the sum of squared residuals.
- A residual is observed minus fitted charges.
- The slope is the estimated difference in mean charges associated with an additional **$1,000 of income**.
- The intercept corresponds to zero income; this lies outside the observed income range and has limited business meaning.
- The slope t test concerns $H_0 : \beta_1 = 0$ under the model assumptions.

Multiple regression

$$Y_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{household size}_i + \varepsilon_i .$$

```
multiple_model <- lm(card_charges ~ income_thousands + household_size, data = consumer)
summary(multiple_model)
```

```
Call:
lm(formula = card_charges ~ income_thousands + household_size,
    data = consumer)

Residuals:
     Min       1Q   Median       3Q      Max
-1180.62  -155.31     7.05   194.56  1309.66

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      1304.905    197.655   6.602 3.29e-08 ***
income_thousands   33.133      3.968   8.350 7.68e-11 ***
household_size    356.296     33.201  10.732 3.12e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 398.1 on 47 degrees of freedom
Multiple R-squared:  0.8256,    Adjusted R-squared:  0.8181
F-statistic: 111.2 on 2 and 47 DF,  p-value: < 2.2e-16
```

```
knitr::kable(confint(multiple_model), digits = 2)
```

	2.5 %	97.5 %
(Intercept)	907.27	1702.54
income_thousands	25.15	41.12
household_size	289.50	423.09

```
anova(simple_model, multiple_model)
```

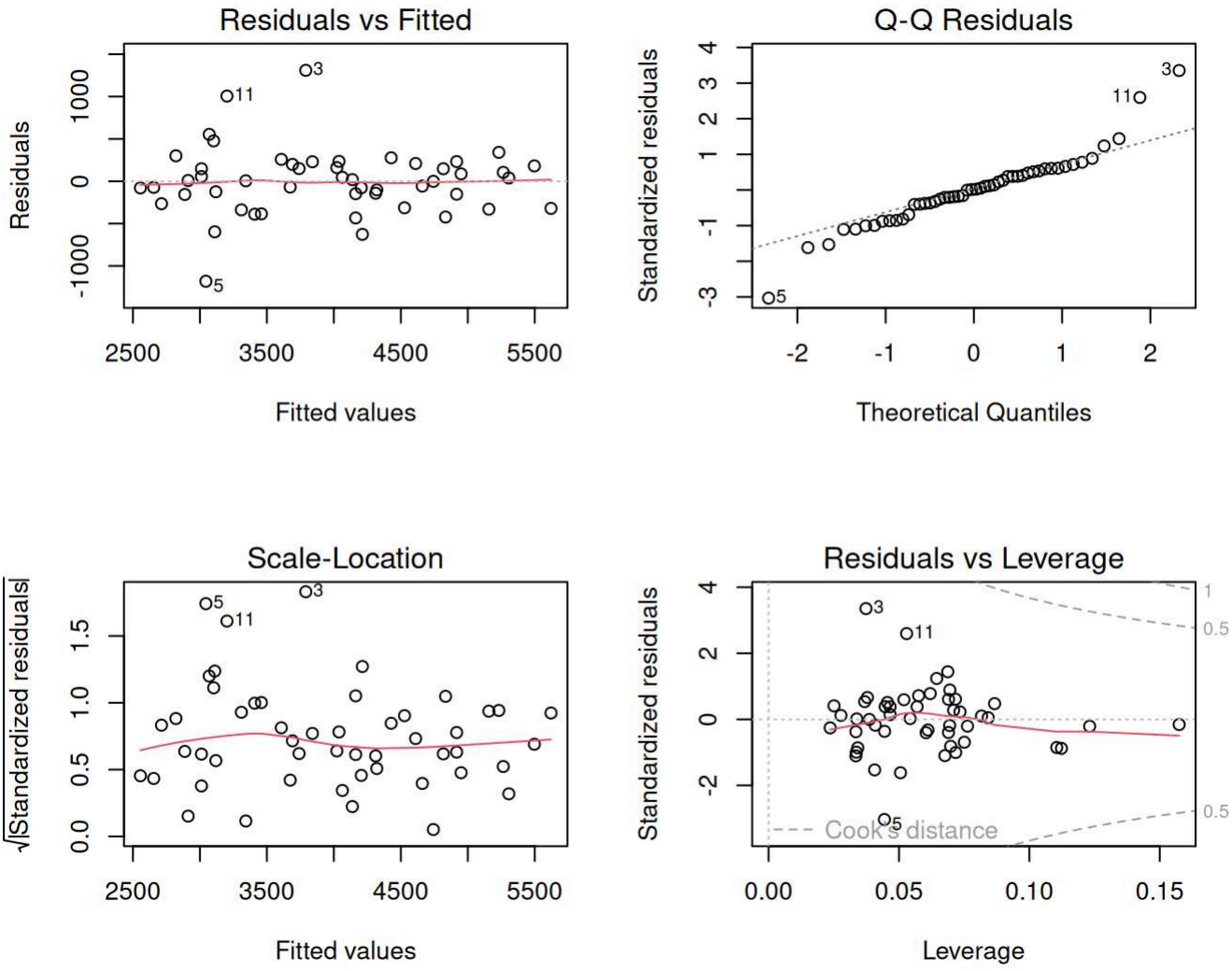
```
Analysis of Variance Table

Model 1: card_charges ~ income_thousands
Model 2: card_charges ~ income_thousands + household_size
  Res.Df      RSS Df Sum of Sq      F    Pr(>F)
1     48 25699404
2     47  7448393  1  18251011 115.17 3.123e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Holding household size fixed, an additional $1,000 of income is associated with about **$33.13** more charges.
- Holding income fixed, one additional household member is associated with about **$356.30** more charges.
- Adjusted R^2 is about **0.818**; this describes fit in the supplied data, not predictive accuracy on new households.
- The nested-model F test asks whether adding household size improves fit beyond income alone, under the model assumptions.
- Statistical adjustment does not turn this observational dataset into a randomized experiment.

Check the model

```r
par(mfrow = c(2, 2))
plot(multiple_model)
```



Residual patterns, normality, scale and influential observations for the multiple regression.

```
par(mfrow = c(1, 1))
```

- Residuals versus fitted: look for curvature or changing spread.
- Q–Q: assess whether normal-error inference is plausible.
- Scale-location: look for unequal residual variation.
- Leverage/Cook's distance: investigate influential records without automatically deleting them.
- Independence comes from collection design; residual plots cannot establish it.
- Classical t/F intervals assume a suitable linear mean, independent errors and constant variance, with normal errors for exact small-sample inference.
- If these assumptions are doubtful, treat the fit as descriptive or justify an alternative analysis before relying on p-values.

Confidence intervals versus prediction intervals

Profile: income of $45,000 and a three-person household, within the observed ranges.

```
profile <- data.frame(income_thousands = 45, household_size = 3)
predict(multiple_model, newdata = profile, interval = "confidence")
```

```
       fit      lwr      upr
1 3864.778 3746.969 3982.587
```

```r
predict(multiple_model, newdata = profile, interval = "prediction")
```

```
       fit      lwr      upr
1 3864.778 3055.303 4674.252
```

- Confidence interval: uncertainty about **mean charges** at that profile.
- Prediction interval: uncertainty for **one new comparable record**; includes individual variation and is wider.
- Neither is a causal estimate of what happens if a household's income changes.
- Avoid extrapolating beyond the observed income and household-size ranges.
- A business deployment would need new-data validation and a defensible target population.

3.10 Your inferential business briefing

A. Sampling audit

1. Draw a new random sample of 100 Superstore orders with a seed you report.
2. Estimate mean order sales and construct a 95% interval with the finite-frame correction.
3. Compare it with the known full-frame mean.
4. Explain why another sample can produce a different interval.
5. State why the result does not describe all US retailers.

B. Display decision

1. Report the COLA difference, interval and p-value in cases per week.
2. Explain why the groups are independent and why random assignment matters.
3. Identify the missing financial information needed before recommending the display.
4. Distinguish the experimental effect from generalization to all supermarkets.

C. Consumer model

1. Fit a simple model using household size alone.
2. Compare it with the two-predictor model using fit and residual evidence.
3. Predict charges for a $40,000-income, four-person household.
4. Explain the difference between a mean-response interval and an individual prediction interval.
5. Write a recommendation containing no causal claim.

```r
# Use the worked analyses as patterns; justify each choice and target.
# Report effects and intervals rather than submitting raw output alone.
```

Extend the methods to another business case

- Oxford Cereal: sampling variation, a mean test and an interval from reported summary inputs.

- BLK Retail Display: follow up the COLA example with a commercial sensitivity briefing.
- Springville Herald, Section B: marketing regression; optional Section C introduces forecasting.
- Perfect Parachute: extend group comparisons to four suppliers with an ANOVA primer.
- Mountain States Potato: multiple regression, diagnostics and production recommendations.
- Harnswell Quality: optional process-control extension with an introductory chart guide.
- Select cases as assigned; the six complete notebooks are collected under Extended Business Cases.

Sources and continuation

- Superstore: Tableau sample data and transaction detail.
- COLA: Berenson, Levine and Krehbiel, *Basic Business Statistics*, ninth edition, pp. 336–339; publisher `COLA.XLS`.
- Consumer: Anderson/Sweeney/Williams data collection, `Consumer.xls`; related Consumer Research case indexed in *Essentials of Statistics for Business and Economics*, ninth edition, pp. 790–791. Local edition details are distinguished in the data guide.
- Probability, estimation, testing and regression follow the sequence of the instructor's original 307102 inferential decks. The current explanations and business analyses are revised for 307312 Applied Statistics.
- Return to the course home page for all practicals and downloads.